DuckDB from the Command Line

Windows: install DuckDB and run your first queries

OMIS 105 - Fall 2026

DuckDB **2.0** or newer is required for this course.

Reviewed November 11, 2026 for DuckDB 2.0. Start here, then follow the slides in order.

1. What You Will Learn

1. Install the standalone DuckDB command-line program.
2. Fix “command not found” or “not recognized.”
3. Open the tutorial folder and query its CSV file.
4. Save a database, change output format, and run a SQL file.
5. Diagnose file, permission, and session errors.

Keep this guide and `data/students.csv` together. You will create `report.sql` during the lesson.

Follow the main lesson first. Use **Help Only If Needed** at the end only when a step fails. You do not have to memorize troubleshooting.

2. Four Words Used in This Guide

Word	Meaning here
Terminal / shell	The terminal is the window; the shell reads your commands.
CLI	“Command-line interface”: the DuckDB program used in this lesson.
Path	A file or folder's location on your computer.
PATH	The list of folders searched when you type a program name.

Enter and **Return** mean the same key in these instructions.

After each command, wait for the result before typing the next one.

3. Two Prompts, Two Kinds of Commands

Where you are	Examples to enter there
Terminal shell	`duckdb --version`, `duckdb`, `cd`
DuckDB prompt (`D`)	`SELECT 42;`, `.tables`, `.quit`

The prompt is a label: **do not type `D`, `$`, `%`, or `PS>`**.
Copy command blocks exactly. Blocks labeled “output” show results;
do not type those results.

If you see `D`, enter `.quit` before a terminal command.
If you see `>>>`, you are in Python: enter `exit()` first.

4. Before Installing

The **CLI** is the DuckDB program you run in a terminal. Having DuckDB work in Python or Marimo does not mean this program is installed.

Open **Command Prompt**: press the Windows key, type `cmd`, and press Enter. If you already use PowerShell, that also works. Then run:

```
duckdb --version
```

The next slide explains what the result means.

4. Before Installing (cont.)

If the version starts with `v2.` (for example `v2.0.0`), DuckDB is installed and your terminal can find it. Skip installation and go to **Open the Tutorial Folder**. If it starts with `v1.`, upgrade it: this course requires DuckDB 2.0, and a 1.x program cannot open database files made by DuckDB 2.0. To upgrade, remove it (if WinGet installed it: `winget uninstall --id DuckDB.cli --exact`), then use Option B.
If you see “command not found” or “not recognized,” follow the installation steps next. Choose only one method. For another error, use the help section.

No Python environment, database server, or DuckDB account is needed.

5. Windows: Choose Your Shell

For a first attempt, use **Command Prompt**: press the Windows key, type `cmd`, and press Enter. If you already use PowerShell, follow the blocks labeled **PowerShell** instead.

Commands labeled PowerShell and Command Prompt are different. Do not use WSL, Git Bash, or a Python prompt for these steps.

Try `duckdb --version`. If not recognized, use the manual route (Option B) — WinGet does not offer DuckDB 2.0 yet. Windows requires the matching Microsoft Visual C++ Redistributable.

6. Windows Option A: WinGet (not yet)

As of November 2026, WinGet's `DuckDB.cli` package stops at **1.5.6** —
not the required **2.0**. **Use Option B for now.**

You can check whether WinGet has caught up (in either shell):

```
winget show --id DuckDB.cli --exact
```

Only if the `Version:` line starts with **2.** may you install with
`winget install --id DuckDB.cli --exact`, close and reopen the terminal,
and confirm `duckdb --version` starts with `v2.`. Then skip Option B and
go to **Windows: Open the Tutorial Folder**.

7. Windows Option B: Download and Extract

1. In **Settings > System > About**, check **System type**: x64 (Intel/AMD) or Arm.
2. At duckdb.org/install, select **Windows / CLI / current stable** (2.0 or newer) and download **duckdb-cli-windows-amd64.tar.gz** (x64) or **duckdb-cli-windows-arm64.tar.gz** (Arm) to your **Downloads** folder.

Since 2.0 the download is a `.tar.gz`, not a `.zip`: on Windows 10, **Extract All** does not open it. The next slide extracts it with Windows' built-in `tar` command, which works on Windows 10 and 11.

7. Windows Option B: Download and Extract (cont.)

Command Prompt (Arm PCs: change `amd64` to `arm64`):

```
mkdir "%USERPROFILE%\duckdb-cli"
tar -xzf "%USERPROFILE%\Downloads\duckdb-cli-windows-amd64.tar.gz" -C "%USERPROFILE%\duckdb-cli"
```

PowerShell:

```
New-Item -ItemType Directory -Force "$env:USERPROFILE\duckdb-cli"
tar -xzf "$env:USERPROFILE\Downloads\duckdb-cli-windows-amd64.tar.gz" -C "$env:USERPROFILE\duckdb-cli"
```

The next slide expects `duckdb-cli\duckdb.exe`. If extraction made an extra folder, move **duckdb.exe** up into `duckdb-cli`. A user-owned folder avoids needing permission to create `C:\duckdb`.

8. Windows: Test Before Editing PATH

PowerShell:

```
Test-Path "$env:USERPROFILE\duckdb-cli\duckdb.exe"
& "$env:USERPROFILE\duckdb-cli\duckdb.exe" --version
```

Command Prompt:

```
dir "%USERPROFILE%\duckdb-cli\duckdb.exe"
"%USERPROFILE%\duckdb-cli\duckdb.exe" --version
```

`Test-Path` should return **True**; `dir` should list **duckdb.exe**.

If not, fix the file location first. The next command should print a version.

PowerShell needs `&` to run a program whose path is in quotes.

9. Windows: Make duckdb Work by Name

For the **manual route**, add the folder containing `duckdb.exe` to PATH:

1. Start menu: search **Edit environment variables for your account**.
2. Under **User variables**, edit **Path**; create it if absent.
3. Add a **new entry**: the full path to your `duckdb-cli` folder.
   Copy it from File Explorer. Do not include `duckdb.exe` or quote marks.
4. Keep existing entries. Do not replace the entire Path value.
5. Close and reopen Windows Terminal, PowerShell, Command Prompt, and VS Code as applicable. Then run `duckdb --version`.

If editing PATH is blocked, use the full-path command from the previous page. `%USERPROFILE%` and `$env:USERPROFILE` mean your home folder; copy them exactly when shown in commands.

10. Windows: Open the Tutorial Folder

Extract the **whole tutorial folder**, including `data/students.csv`.
In File Explorer, open that folder, type **cmd** in its address bar, and
press Enter. A Command Prompt opens at the correct location. Run:

```
cd
dir data\students.csv
```

If using PowerShell instead, use **Copy as path** on the folder and enter
`Set-Location "the copied full folder path"`, then `Get-Location`.
The quoted path is a placeholder: replace it with your actual folder.
It may include **OneDrive**; copy the real location from File Explorer.

11. Windows: Paths Differ Between Shells

To navigate manually, replace the example path with your real location:

Command Prompt:

```
cd /d "C:\Users\maria\Documents\DuckDB_from_Command_Line"
```

PowerShell:

```
Set-Location "C:\Users\maria\Documents\DuckDB_from_Command_Line"
```

`/d` lets Command Prompt change drives too. `%USERPROFILE%` is CMD syntax; `$env:USERPROFILE` is PowerShell syntax. Bare `cd` reports the folder in CMD but changes to the home folder in PowerShell.

12. Checkpoint: Does DuckDB Run?

In your terminal:

```
duckdb --version
duckdb -c "SELECT 6 * 7 AS answer;"
```

The first command should print a version number (`v2.` recommended); the second should return **42**. The version line looks like this — the code at the end differs by build (do not type this):

```
v2.0.0 (Cyanoptera) …
```

If typing `duckdb` still fails, see **Help Only If Needed** at the end.
It explains how to start DuckDB using the location of its program file.

13. Checkpoint: Can It Read the Sample?

In your terminal, in the tutorial folder:

```
duckdb -c "SELECT COUNT(*) AS students FROM read_csv('data/students.csv');"
```

Expect **8**. If this fails but the 42 check worked, installation is OK:
check your current folder and the CSV filename.

The CSV should begin `id,name,major,gpa`. A file containing HTML is a saved web page, not the dataset. Download the actual file or whole repo.

Once both checkpoints pass, proceed with the SQL lesson.

14. Start DuckDB and Type a Query

In your terminal:

```
duckdb
```

When you see **D**, enter this SQL and press Enter/Return:

```
SELECT 6 * 7 AS answer;
```

You should see **42**. Startup text and spacing vary by version.

This session is **in-memory**: its database tables disappear on exit.

Writing a separate file is different; exported files remain on disk.

15. Finish or Cancel an Incomplete Command

SQL statements finish with a **semicolon** (`;`). DuckDB may show a continuation prompt if a statement, quote, or parenthesis is incomplete.

- For a missing semicolon, type `;` and press Enter.
- If you pasted the wrong text or have unmatched quotes, press **Ctrl+C** to cancel the pending input, then re-enter the complete command.
- Dot-commands such as `.quit` must start on their own fresh line. They have **no semicolon**.

A continuation prompt does not mean every problem is a missing `;` .

16. Query the CSV Directly

At the DuckDB prompt, run each block separately:

```
SELECT * FROM read_csv('data/students.csv');
```

```
SELECT major, ROUND(AVG(gpa), 2) AS avg_gpa
FROM read_csv('data/students.csv')
GROUP BY major
ORDER BY avg_gpa DESC, major;
```

No import step is needed. This reads the CSV; it does not create a stored table named `students`. DuckDB looks for `data/` inside the folder from which you started the program.

17. Expected GPA Results

major	avg_gpa
Information Systems	3.83
Accounting	3.50
Finance	3.35
Marketing	3.35

Finance and Marketing tie; sorting by `major` breaks the tie consistently.

DuckDB may display `3.5` rather than `3.50`; they are the same number.

The default **duckbox** format may also show column types.

18. Create a Table for This Session

At the DuckDB prompt:

```
CREATE TABLE students AS
SELECT * FROM read_csv('data/students.csv');
```

Then enter these **one line at a time**:

```
.tables
.schema students
```

`.tables` lists stored tables/views, not every CSV in the folder.

If `students` already exists, query it instead of running CREATE again.

19. Change How Results Look

At the DuckDB prompt, one line at a time:

```
.mode csv
.headers on
SELECT id, name FROM students ORDER BY id LIMIT 2;
```

Expected CSV output:

```
id,name
1,Alice
2,Bob
```

Other modes: `.mode markdown`, `.mode json`, `.mode table`.

Use `.mode duckbox` to return to the default display.

20. Headers and Timing

At the DuckDB prompt, one line at a time:

```
.mode csv
.headers off
SELECT COUNT(*) FROM students;
.headers on
.timer on
SELECT COUNT(*) FROM students;
.timer off
.mode duckbox
```

The first count is **8** without a header. The second adds a header and timing information. Timing values vary; these settings persist in the session.

21. Save a Database and Reopen It

First exit the in-memory session with `.quit`. **In your terminal:**

```
duckdb class_cli.db
```

At D, create the table once in this file database:

```
CREATE TABLE students AS
SELECT * FROM read_csv('data/students.csv');
```

Enter `.quit`, then run `duckdb class_cli.db` again in the same folder.
At D, `SELECT COUNT(*) FROM students;` should still return **8**.
If the table already exists, skip CREATE and query it.

22. Keep Track of Your Saved Database

- A filename such as `class_cli.db` refers to a file in the current folder. Starting from a different folder can create a different, empty database.
- Use `.databases` at D to inspect the open database; use `.tables` to inspect its tables. Quote terminal paths containing spaces.
- A database's parent folder must already exist and be writable.
- `.open file.db` switches databases; it does **not** save/copy your previous in-memory tables into the file.
- The statements in this lesson save automatically in a file database. If your instructor has you use `BEGIN`, use `COMMIT` before exiting.

23. Create a Script File

In a plain-text editor, save these lines as **report.sql** in the tutorial folder. Use VS Code or Notepad (Save as type: **All files**).
Check that the file is named `report.sql`, not `report.sql.txt`.

```
SELECT COUNT(*) AS students FROM read_csv('data/students.csv');
SELECT major, ROUND(AVG(gpa), 2) AS avg_gpa
FROM read_csv('data/students.csv')
GROUP BY major
ORDER BY avg_gpa DESC, major;
```

24. Run Your Script File

After saving **report.sql**, exit DuckDB with `.quit` first.

In the terminal, from this tutorial folder:

```
duckdb -bail -f report.sql
```

Expect the row count **8**, then the four GPA results shown earlier.

`-bail` stops if an error occurs. `-f` runs the SQL file and exits.

Alternatively, if you have not exited DuckDB, enter `.read report.sql` at D. In either case, start DuckDB from the tutorial folder so it can find `data/students.csv`.

25. Optional Practice: Write Another SQL File

Use VS Code or Notepad as your plain-text editor.

Save this as **my_query.sql** in the tutorial folder:

```
SELECT major, COUNT(*) AS n
FROM read_csv('data/students.csv')
GROUP BY major
ORDER BY major;
```

Exit DuckDB first, then run `duckdb -bail -f my_query.sql` in the shell.

Check it is not `my_query.sql.txt` or rich text. In Notepad, choose **All files**. Use straight quotes, not smart/curly quotes.

26. Quotes, Spaces, and Copy/Paste

For simple one-off SQL, double-quote the shell argument and single-quote SQL strings:

```
duckdb -c "SELECT 'Finance' AS major;"
```

SQL uses single quotes for text values; double quotes are for names, such as column names.
The terminal may treat `$`, `!`, and some quotes as special instructions.
For SQL containing these characters or multiple lines, use a `.sql` file.

Copy commands from Markdown when possible. PDF line wraps, curly quotes, and copied prompt labels can alter a command. Enter one block at a time.

27. Practice: Seven Checks

1. In the shell: run `duckdb --version` and the 42 checkpoint.
   Expect a version number and the answer **42**.
2. Confirm you are in this tutorial's folder.
3. In the shell: run the CSV checkpoint and confirm the count is **8**.
4. Run `duckdb` ; at D, run the average-GPA query.
5. Try `.mode markdown` , then rerun the query.
6. Create `report.sql` as shown earlier. Enter `.quit` , then run `duckdb -bail -f report.sql` in the shell.
7. Create/reopen `class_cli.db` and verify its table persists.

If typing `duckdb` fails, use **Help Only If Needed** below.
First get the program working, then check the location of your data.

28. Help Only If Needed

If you completed all seven checks and saw the expected results, you have finished the lesson.
Use the following pages only for the problem you are seeing:

- `duckdb` is not found or not recognized.
- The program exists but cannot open.
- A CSV, SQL file, or saved table cannot be found.
- The database is locked, or the display looks unexpected.

Try the matching fix. If it does not help, use **Still Stuck?** at the end to send the instructor the details needed to help you.

29. Windows: “duckdb is not recognized”

In **either shell**:

```
where.exe duckdb
duckdb --version
```

If not found: verify installation, restart the terminal application, and
test the full executable path. Reopening alone cannot add a missing entry.

If you once installed DuckDB with WinGet, `winget list --id DuckDB.cli --exact`
shows it; it is 1.x (the course requires 2.0) — to upgrade, uninstall it and use the manual route,
so that the old copy is not found first.
For manual installation, verify the folder and User Path entry.

In PowerShell, `Get-Command duckdb -All` also reveals aliases or multiple
copies. Bare `where` is a PowerShell alias, so use **where.exe**.

30. Windows: Continue Without Permanent PATH

For the manual installation, add the folder for **this session only**:

PowerShell:

```
$env:Path = "$env:USERPROFILE\duckdb-cli;$env:Path"
duckdb --version
```

Command Prompt:

```
set "PATH=%USERPROFILE%\duckdb-cli;%PATH%"
duckdb --version
```

Use the actual folder you installed into. These settings disappear when the shell closes; the full-path commands also work without changing PATH.

31. Windows: File Exists but Will Not Run

- **In its folder, PowerShell cannot find duckdb.exe:** use `.\duckdb.exe --version`; PowerShell does not search the current folder implicitly. Running it this way does not add it to PATH.
- **VCRUNTIME / MSVCP DLL missing:** install or repair the matching Microsoft Visual C++ Redistributable from Microsoft's site.
- **This app cannot run on your PC:** verify architecture (amd64 vs arm64 download), and check OS support. Do not download random DLLs.
- **Access denied / file disappears:** check device security notices; ask IT if execution or extraction is blocked.

32. Windows: SmartScreen and Managed PCs

If you see **Windows protected your PC**, confirm the executable came from DuckDB's official installation page.

Use **More info > Run anyway** only for a trusted download and when your device policy permits it. If that option is absent, contact IT.
Do not disable security software or assume WinGet bypasses policy.

For school PCs, a full-path command solves PATH lookup issues, but cannot solve a blocked executable or missing runtime dependency.

33. Windows: Continue Without Fixing PATH First

Use the working manual-install path instead of the command name:

PowerShell:

```
& "$env:USERPROFILE\duckdb-cli\duckdb.exe" -c "SELECT 42;"
```

Command Prompt:

```
"%USERPROFILE%\duckdb-cli\duckdb.exe" -c "SELECT 42;"
```

To open an interactive session, omit `-c "SELECT 42;"`.
Use your actual installation folder if different. This avoids PATH lookup problems; it cannot fix a blocked program or missing Windows component.

34. Recovery: CSV or SQL File Not Found

1. If you are at D, press Ctrl+C only if input is unfinished; then enter `.quit`.
2. In the shell, check the current folder and list the actual filename.
3. Change into the folder containing **data/**. For scripts, also check that you created **report.sql** there as instructed.
4. Retry the eight-row checkpoint. If your problem was with the script, also retry `duckdb –bail –f report.sql` after confirming the file exists.

If using a synced folder (OneDrive/iCloud), download files locally first.
A full path can avoid ambiguity; use single quotes for paths in SQL.
If a path contains an apostrophe, double it inside the SQL string.

A `.csv` extension alone does not prove the file is valid CSV.

35. Recovery: Table, Lock, or Write Errors

- **Table does not exist:** you may be in a new in-memory session or the wrong database. Query the CSV directly or reopen the right `.db`.
- **Table already exists:** reuse it; do not drop it just to clear the error.
- **Could not set lock:** close other CLI sessions or Marimo/Python connections using that file, then retry. Do not delete lock/WAL files.
- **Read-only / cannot open:** check the parent folder, permissions, disk space, and whether you opened with `-readonly`.

Prefer a local writable folder for the database. **Version errors:** a `.db` file saved by DuckDB 2.0 cannot be opened by DuckDB 1.x (an older CLI, Python package, or qStudio) — upgrade that program; never overwrite the database to get rid of the error.

36. Recovery: Display or Startup Problems

- Garbled table borders? At D, try `.mode table` for ASCII borders. `.mode ascii` means a different machine-oriented format.
- Results look different? Check `.show`; reset with `.mode duckbox` and `.headers on`. A startup `.duckdbrc` can customize settings.
- To bypass it, first save a plain-text **clean_start.sql** containing only `-- No startup settings`. Save it in the tutorial folder, exit DuckDB with `.quit`, and run this command from that folder:

```
duckdb -init clean_start.sql -c "SELECT 42 AS answer;"
```

Do not delete personal startup files. Different colors or layouts do not change query results.

37. Other File and Download Problems

- SQL files with spaces in their names need quotes. In the terminal: `duckdb -bail -f "my reports/report.sql"`.
- A failed download may be caused by a school network, proxy, or certificate restriction. Ask IT; do not disable certificate checks.
- If the computer or operating system is unsupported, ask the instructor for an approved alternative computer.
- If you have not created `report.sql` or `clean_start.sql`, follow the earlier instructions before trying to run either file. They are not supplied.

38. Still Stuck? Send Useful Details

Send the instructor:

- Your Windows version and shell (PowerShell or Command Prompt).
- The exact command and complete error text.
- Whether a full-path `--version` command works.
- Your current folder and whether `data/students.csv` exists.
- `duckdb --version`, if available; redact private path information.

Use the help pages in this guide and the official references below.
Check `.help` and `duckdb -help` for the options of your installed version.

39. Official References and Validation

- DuckDB installation
- DuckDB CLI overview
- CLI arguments
- Dot-commands
- Output formats

40. Windows References and Validation

- Microsoft: WinGet and App Installer
- PowerShell: command resolution
- Microsoft Visual C++ Redistributable

Windows setup was checked against official documentation (download names on duckdb.org/install and WinGet's package list, 2026-10-25), not tested on a Windows PC. Every SQL example and its expected result was re-checked with DuckDB 2.0.0; the CLI-only parts were re-tested with CLI 2.0.0 on macOS (2026-11-11).